Sports Science Research Update

August 2023

A case report of reversible cognitive decline in a former professional American-style football player: Findings from the Football Players Health Study in-person assessments. Current Sports Medicine Reports, Vol. 22(5): 154-157, 2023.

At-risk tackling techniques and effectiveness in NCAA Division I collegiate football. The Orthopaedic Journal of Sports Medicine, 11(7): 1-6, 2023.

Beneficial effects of oral and topical sodium bicarbonate during a battery of team sport-specific exercise tests in recreationally trained male athletes. Journal of the International Society of Sports Nutrition, Ahead of Print: 1-21, 2023.

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Editorial – Psychological readiness to return to sport: Fear of reinjury is the leading reason for failure to return to competitive sport and is modifiable. Arthroscopy: The Journal of Arthroscopic and Related Surgery, Vol. 39(8): 1775-1778, 2023.

Sleep/wake disorders after sports concussion: Risks, revelations, and interventions. Journal of Clinical Neurophysiology, Vol. 40(5): 417-425, 2023.

Sports Science Research Update

A case report of reversible cognitive decline in a former professional American-style football player: Findings from the Football Players Health Study in-person assessments. Current Sports Medicine Reports, Vol. 22(5): 154-157, 2023.

Introduction - American-style football (ASF) is a popular sport that has a high risk for injury. Both media and player attention have increased interest in understanding the health consequences of participation in the sport, including both the short- and long-term contributions of repetitive head trauma to the development of neurodegenerative disease. Frontotemporal lobal degeneration has been seen in concert with chronic traumatic encephalopathy (CTE) diagnoses antemortem (1). Recognizing gaps in diagnosis of CTE, experts have developed research diagnostic criteria for CTE including head trauma, clinical features, progressive course and absence of alternative medical, and psychological or neurological diagnosis (2). In a clinic-pathological case series of patients with a history of repetitive head injuries and traumatic encephalopathy syndrome, researchers found that two of the nine donated brains exhibited frontotemporal lobar degeneration alongside reported memory and executive function decline (3). In addition, frontotemporal atrophy was reported to be more common in football players who had severe CTE in comparison with their counterparts with more moderate or no CTE (3–7). Findings from several studies investigating the relationship between repetitive subconcussive or concussive injuries and clinical neurological outcomes have been inconclusive, and researchers are still unsure if participation in collision sports has cumulative neurological effects over the course of an athlete's career (8,9). Despite the documented histopathological changes of frontotemporal atrophy observed in former athletes, Frontotemporal dementia (FTD) diagnoses have not been documented in living former ASF players. FTD is a pathologically and clinically diverse syndrome characterized by impaired judgment, apathy, and other behavioral and/or dramatic personality changes (10). FTD diagnoses continue to remain a challenge among clinicians and researchers, resulting in many patients being misdiagnosed (11). Population studies conducted in Europe and North America report that FTD prevalence ranges from 2.7/100,000 to 31/100,000 (12). Many of the clinical symptoms of neurodegenerative dis ease in former ASF players may overlap with other potentially treatable medical conditions. Understanding methods to optimize treatment of the postcareer health in professional athletes is a growing interest in the sports medicine community. This report presents a case of a middle-aged former professional ASF player who was suffering from life-altering cognitive, behavioral, and personality changes that were originally diagnosed as FTD. Upon completion of a clinical research study, this diagnosis was challenged by alternative findings that better explained his neurocognitive issues. Recognition and treatment of hydrocephalus in the former athlete resulted in improvement of his overall health.

Sports Science Research Update

At-risk tackling techniques and effectiveness in NCAA Division I collegiate football. The Orthopaedic Journal of Sports Medicine, 11(7): 1-6, 2023.

Background: Head-down tackling has been associated with injuries to the brachial plexus, cervical spine, and head in high school and collegiate American football. Head-down tackling has also been associated with decreased effectiveness in successful tackles compared with head-up tackling.

Purpose: To assess tackling techniques used during National Collegiate Athletic Association (NCAA) Division I football games and to evaluate the successful tackling rates according to technique. Study

Design: Cross-sectional study.

Methods: Three reviewers analyzed 1000 consecutive solo defensive tackling attempts made in the 2021 season (October to December) by 8 universities within the NCAA Southeastern Conference. Slow-motion replays were used to analyze the success of the tackling attempt, the tackling method, and the initial point of contact with respect to the offensive player's waist. The chi-square or Fisher exact test was used to analyze categorical data, and the 2-tailed Student t-test or the Mann-Whitney U test was used to analyze continuous data.

Results: Head-up and head-down tackling occurred in 902 (90.2%) and 98 (9.8%) tackle attempts, respectively. Head-up tackles were successful in 76.2% of the attempts compared with 55.1% for head-down tackles ($P < .001$). Tackles were made at or above the offensive player's waist in 777 (77.7%) attempts and below the waist in 223 (22.3%) attempts. Tackles at or above the waist were successful in 77.6% of the attempts compared with 61.9% of tackles below the waist ($P < .001$). The inside-shoulder method was used in 592 (59.2%) tackles, the arm method in 317 (31.7%), the head-across-the-bow method in 72 (7.2%), and the helmet-to-helmet method in 19 (1.9%). Inside-shoulder tackles had the highest success rate of 93.2%, compared with 41.6% for arm ($P < .001$), 59.7% for head-across-the-bow ($P < .001$), and 73.7% for helmet-to-helmet ($P = .001$) tackles. Inside-shoulder tackles resulted in head-up tackling in 92.9% compared with 41.7% for head-across-the-bow ($P < .001$) and 57.9% for helmet-to-helmet ($P < .001$) tackles. There were no recorded injuries to the tackler.

Conclusion: Head-up tackles, tackles made at or above the offensive player's waist, and inside-shoulder tackles had the highest success rates. Head-down tackling and tackling below the waist were associated with poor tackling methods, including head-across-the-bow and helmet-to-helmet tackles, which had lower success rates.

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Beneficial effects of oral and topical sodium bicarbonate during a battery of team sport-specific exercise tests in recreationally trained male athletes. Journal of the International Society of Sports Nutrition, Ahead of Print: 1-21, 2023.

Objective: This study examined the effects of oral and topical (PR Lotion; Momentous) sodium bicarbonate (NaHCO3) during a battery of team sport-specific exercise tests.

Method: In a block randomized, crossover, double-blind, placebo controlled design, 14 recreationally trained male team sport athletes performed a familiarization visit and three experimental trials receiving: (i) 0.3 g·kg−1 body mass (BM) NaHCO3 in capsules + placebo lotion (SB-ORAL), (ii) placebo capsules +0.9036 g·kg−1 BM PR Lotion (SB-LOTION), or (iii) placebo capsules + placebo lotion (PLA). Supplements were given ~120 min prior to the team sport-specific exercise tests: countermovement jumps (CMJ), 8 × 25 m repeated sprints and Yo-Yo Intermittent Recovery Level 2 (Yo-Yo IR2). Blood acid–base balance (pH, bicarbonate) and electrolytes (sodium, potassium) were measured throughout. Rating of perceived exertion (RPE) was recorded after each sprint and post-Yo-Yo IR2.

Results: Distance covered during the Yo-Yo IR2 was 21% greater for SB-ORAL compared with PLA (+94 m; $p = 0.009$, $d = 0.64$) whereas performance was only 7% greater for SB-LOTION compared with PLA (480 ± 122 vs. 449 ± 110 m; $p = 0.084$). Total completion time for the 8 × 25 m repeated sprint test was 1.9% faster for SB-ORAL compared with PLA (−0.61 s; $p = 0.020$, $d = 0.38$) and 2.0% faster for SB-LOTION compared with PLA (−0.64 s; $p = 0.036$, $d = 0.34$). CMJ performance was similar between treatments ($p > 0.05$). Blood acid– base balance and electrolytes were significantly improved for SB-ORAL compared with PLA, but no differences were observed for SB-LOTION. Compared to PLA, RPE was lower for SB-LOTION after the fifth ($p = 0.036$), sixth ($p = 0.012$), and eighth ($p = 0.040$) sprints and for SB-ORAL after the sixth ($p = 0.039$) sprint.

Conclusions: Oral NaHCO3 improved 8 × 25 m repeated sprint (~2%) and Yo-Yo IR2 performance (21%). Similar improvements in repeated sprint times were observed for topical NaHCO3 (~2%), but no significant benefits were reported for Yo-Yo IR2 distance or blood acid–base balance compared to PLA. These findings suggest that PR Lotion might not be an effective delivery system for transporting NaHCO3 molecules across the skin and into systematic circulation, therefore further research is needed to elucidate the physiological mechanisms responsible for the ergogenic effects of PR Lotion.

Sports Science Research Update

Cannabidiol for musculoskeletal regenerative medicine. Experimental Biology and Medicine, 248: 445-455, 2023.

Abstract - Chronic musculoskeletal (MSK) pain is one of the most prevalent causes, which lead patients to a physician's office. The most common disorders affecting MSK structures are osteoarthritis, rheumatoid arthritis, back pain, and myofascial pain syndrome, which are all responsible for major pain and physical disability. Although there are many known management strategies currently in practice, phytotherapeutic compounds have recently begun to rise in the medical community, especially cannabidiol (CBD). This natural, non-intoxicating molecule derived from the cannabis plant has shown interesting results in many preclinical studies and some clinical settings. CBD plays vital roles in human health that go well beyond the classic immunomodulatory, anti-inflammatory, and antinociceptive properties. Recent studies demonstrated that CBD also improves cell proliferation and migration, especially in mesenchymal stem cells (MSCs). The foremost objective of this review article is to discuss the therapeutic potential of CBD in the context of MSK regenerative medicine. Numerous studies listed in the literature indicate that CBD possesses a significant capacity to modulate mammalian tissue to attenuate and reverse the notorious hallmarks of chronic musculoskeletal disorders (MSDs). The most of the research included in this review report common findings like immunomodulation and stimulation of cell activity associated with tissue regeneration, especially in human MSCs. CBD is considered safe and well tolerated as no serious adverse effects were reported. CBD promotes many positive effects which can manage detrimental alterations brought on by chronic MSDs. Since the application of CBD for MSK health is still undergoing expansion, additional randomized clinical trials are warranted to further clarify its efficacy and to understand its cellular mechanisms.

Impact Statement - Cannabidiol (CBD) has gained a lot of significance from orthopedists and sports medicine physicians due to its potential part in the treatment of chronic pain in musculoskeletal (MSK) conditions. CBD plays vital physiological roles in human health that go well beyond immunomodulation, anti-inflammation, and antinociception. Recent investigations show that CBD also enhances cell proliferation and migration, especially in human MSCs. CBD is still relatively new in MSK medicine, and even though new studies are emerging, the clinical application of CBD requires more robust data from clinical trials to further elucidate the mechanisms that contribute to the improvement of MSK structures.

Sports Science Research Update

Chronic traumatic encephalopathy neuropathologic change is uncommon in men who played amateur American football. Frontiers in Neurology, 14(1143882): 1-12, 2023.

Introduction: We examined postmortem brain tissue from men, over the age of 50, for chronic traumatic encephalopathy neuropathologic change (CTE-NC). We hypothesized that (i) a small percentage would have CTE-NC, (ii) those who played American football during their youth would be more likely to have CTE-NC than those who did not play contact or collision sports, and (iii) there would be no association between CTE-NC and suicide as a manner of death.

Methods: Brain tissue from 186 men and accompanying clinical information were obtained from the Lieber Institute for Brain Development. Manner of death was determined by a board-certified forensic pathologist. Information was obtained from next of kin telephone interviews, including medical, social, demographic, family, and psychiatric history. The 2016 and 2021 consensus definitions were used for CTE-NC. Two authors screened all cases, using liberal criteria for identifying "possible" CTE-NC, and five authors examined the 15 selected cases.

Results: The median age at the time of death was 65 years (interquartile range = 57– 75; range = 50–96). There were 25.8% with a history of playing American football and 36.0% who had suicide as their manner of death. No case was rated as definitively having "features" of CTE-NC by all five authors. Ten cases were rated as having features of CTE-NC by three or more authors (5.4% of the sample), including 8.3% of those with a personal history of playing American football and 3.9% of those who did not play contact or collision sports. Of those with mood disorders during life, 5.5% had features of CTE-NC compared to 6.0% of those who did not have a reported mood disorder. Of those with suicide as a manner of death, 6.0% had features of CTE-NC compared to 5.0% of those who did not have suicide as a manner of death.

Discussion: We did not identify a single definitive case of CTE-NC, from the perspective of all raters, and only 5.4% of cases were identified as having possible features of CTE-NC by some raters. CTE-NC was very uncommon in men who played amateur American football, those with mood disorders during life, and those with suicide as a manner of death.

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Commotio Coris in 2023. Sports Medicine, Ahead of Print: 1-10, 2023.

Abstract - Since the nationally televised cardiac arrest of American National Football League player Damar Hamlin in January 2023, commotio cordis has come to the forefront of public attention. Commotio cordis is defined as sudden cardiac arrest due to direct trauma to the precordium resulting in ventricular fibrillation or ventricular tachycardia. While the precise incidence of commotio cordis is not known due to a lack of standardized, mandated reporting, it is the third most common cause of sudden cardiac death in young athletes, with more than 75% of cases occurring during organized and recreational sporting events. Given that survival is closely tied to how quickly victims receive cardiopulmonary resuscitation and defibrillation, it is crucial to raise awareness of commotio cordis so that athletic trainers, coaches, team physicians, and emergency medical personnel can rapidly diagnose and treat this often-fatal condition. Broader distribution of automated external defibrillators in sporting facilities as well as increased presence of medical personnel during sporting events would also likely lead to higher survival rates.

Key Points

- Commotio cordis is defined as sudden cardiac arrest due to blunt trauma to the left chest area directly over the heart, and rapid recognition and intervention is crucial to improve chances of survival.
- This review identified a need for improved access to existing commotio cordis and sudden cardiac death registries, along with more standardized processes for reporting new cases of commotio cordis by athletic trainers, coaches, team physicians, and other healthcare professionals.
- Increased distribution and improved maintenance of automated external defibrillators (AEDs) in sporting facilities as well as dedicated training in the use of AEDs would likely lead to higher survival rates for commotio * cordis as well as other causes of sudden cardiac death

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Consensus Statement on Concussion in Sport: The 6th International Conference on Concussion in Sport – Amsterdam, October 2022. British Journal of Sports Medicine, 57: 695-711, 2023.

Abstract - For over two decades, the Concussion in Sport Group has held meetings and developed five international statements on concussion in sport. This 6th statement summarizes the processes and outcomes of the 6th International Conference on Concussion in Sport held in Amsterdam on 27–30 October 2022 and should be read in conjunction with the (1) methodology paper that outlines the consensus process in detail and (2) 10 systematic reviews that informed the conference outcomes. Over 3½ years, author groups conducted systematic reviews of predetermined priority topics relevant to concussion in sport. The format of the conference, expert panel meetings and workshops to revise or develop new clinical assessment tools, as described in the methodology paper, evolved from previous consensus meetings with several new components. Apart from this consensus statement, the conference process yielded revised tools including the Concussion Recognition Tool-6 (CRT6) and Sport Concussion Assessment Tool-6 (SCAT6, Child SCAT6), as well as a new tool, the Sport Concussion Office Assessment Tool-6 (SCOAT6, Child SCOAT6). This consensus process also integrated new features including a focus on the para-athlete, the athlete's perspective, concussion-specific medical ethics and matters related to both athlete retirement and the potential long-term effects of SRC, including neurodegenerative disease. This statement summarizes evidence-informed principles of concussion prevention, assessment and management, and emphasizes those areas requiring more research.

Key points

⇒ The Amsterdam 2022 International Consensus Statement on Concussion in Sport summarizes published evidence at the time of the conference and should be read together with the 10 systematic reviews and the methodology paper.

⇒ Content and methodological advances were made in the consensus process including anonymous voting, summaries of alternate viewpoints, declarations of conflicts of interest in the open conference, plus inclusion of the athlete voice, para sport considerations and ethical perspectives.

⇒ The Concussion in Sport Group definition of concussion was updated while work continues toward a unified conceptual and operational definition.

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⇒ Sport-specific strategies recommended as concussion prevention interventions include policy or rule changes reducing collisions, neuromuscular training in warm-ups, mouthguard use in ice hockey and implementation of optimal concussion management strategies to reduce recurrent concussion rates.

⇒ The Concussion Recognition Tool-6 (CRT6), Sport Concussion Assessment Tool-6 (SCAT6) and Child SCAT6 provide updated iterations of the acute sport-related concussion (SRC) tools best used in the first 72 hours (and up to 1week) after injury. New office tools, the Sport Concussion Office Assessment Tool-6 (SCOAT6) and Child SCOAT6, were designed to better guide evaluation and management in an office setting from 72 hours after injury and for serial evaluations in the following weeks. The overlap between the SCAT6 and SCOAT6 is intentional and designed to facilitate easy transitions across tools.

⇒ The results of computerized neurocognitive tests should be interpreted in the context of broader clinical findings and are not to be used in isolation to inform management or diagnostic decisions.

⇒ Advanced neuroimaging, fluid-based biomarkers, genetic testing and emerging technologies are valuable research tools for the study of concussion but not yet suited for routine use in clinical practice.

⇒ Return-to-learn and return-to-sport strategies have been updated based on evolving evidence.

⇒ Strong evidence exists regarding the benefits of physical activity and aerobic exercise treatment as early interventions.

⇒ Cervicovestibuar rehabilitation is indicated for athletes with neck pain, headaches, dizziness and/or balance problems.

⇒ Individuals with persisting symptoms (ie, symptom duration >4 weeks) should be evaluated with a multimodal clinical assessment including the use of standardized and validated symptom rating scales.

⇒ The potential long-term effects of SRC and repetitive head impacts are areas of ongoing public health interest and concern among both healthcare professionals and the general public. It is proposed that a working group representing multiple disciplines and perspectives be established to guide appropriate research in this area.

⇒ Decisions regarding retirement or discontinuation from contact or collision sports are complex, multifaceted and should be individualized to consider patient, injury, sport-specific, ethical and psychosocial factors. A comprehensive multidisciplinary clinical evaluation is often necessary to inform decisions.

⇒ Limited evidence exists on SRC in patients aged 5–12 years.

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⇒ Concussion diagnosis and management in para-athletes is challenging with limited data, requiring further research and dedicated clinical recommendations that consider a range of impairments.

⇒ Future research and consensus processes for concussion in sport should continue to evolve with an inclusive and interdisciplinary approach.

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Effects of muscle fatigue on exercise-induced hamstring muscle damage: A three-armed randomized controlled trial. European Journal of Applied Physiology, Ahead of Print: 1-17, 2023.

Purpose: Hamstring injuries in soccer reportedly increase towards the end of the matches' halves as well as with increased match frequency in combination with short rest periods, possibly due to acute or residual fatigue. Therefore, this study aimed to investigate the effects of acute and residual muscle fatigue on exercise-induced hamstring muscle damage.

Methods: A three-armed randomized-controlled trial, including 24 resistance-trained males, was performed allocating subjects to either a training group with acute muscle fatigue+eccentric exercise (AF/ECC); residual muscle fatigue+eccentric exercise (RF/ECC) or a control group with only eccentric exercise (ECC). Muscle stiffness, thickness, contractility, peak torque, range of motion, pain perception, and creatine kinase were assessed as muscle damage markers pre, post, 1 h post, and on the consecutive three days.

Results: Significant group × time interactions were revealed for muscle thickness ($p=0.02$) and muscle contractility parameters radial displacement (Dm) and contraction velocity (Vc) (both $p=0.01$), with larger changes in the ECC group (partial $\eta 2=0.4$). Peak torque dropped by an average of 22% in all groups; stiffness only changed in the RF/ECC group ($p=0.04$). Muscle work during the damage protocol was lower for AF/ECC than for ECC and RF/ECC ($p=0.005$).

Conclusion: Hamstring muscle damage was comparable between the three groups. However, the AF/ECC group resulted in the same amount of muscle damage while accumulating significantly less muscle work during the protocol of the damage exercise.

Sports Science Research Update

Hamstring muscle fibre typology is not associated with hamstring strain injury history or performance in amateur male soccer players: A retrospective magnetic resonance spectroscopy study. Biology of Sport, 40(4): 1177-1186, 2023.

Abstract - Hamstring strain injuries (HSI) are still the most common injuries in soccer. Recent research has been focusing on the role of hamstring muscle morphology and architecture. The hamstring's fibre type composition might play a role as well, but this has never been investigated in the light of HSI risk in an athletic population. The purpose of this study was to investigate the association between hamstring muscle fibre type, hamstring strain injury history (HSIH), performance and isokinetic strength in a population of amateur male soccer players. In this cross-sectional observational study, 44 male soccer players (22 with and 22 without HSIH) participated. The research consisted of a non-invasive fibre composition evaluation using proton magnetic resonance spectroscopy (1 H-MRS), functional performance (evaluated by means of maximal jumping height, maximal sprinting speed and hamstring muscle strength endurance (single leg hamstring bridge testing)), and isokinetic strength testing. The results revealed that hamstring carnosine concentration demonstrated a high inter-individual variability within this soccer population and was not significantly associated with either HSIH or with any of the functional performance parameters. The only secondary outcome measure presenting a significant association with the intramuscular carnosine content was the hamstrings' explosive strength production capacity, objectified by means of the time to peak torque (TPT), measured concentrically at an angular velocity of 240 degrees/second (°/s) during isokinetic strength testing. This TPT was significantly shorter in players presenting higher carnosine concentrations ($p = 0.044$). The findings indicate that in male amateur soccer players (1) the hamstrings have no distinct fibre type dominance and (2) fibre typology in this population does not relate to HSIH or performance.

Sports Science Research Update

"Heads Up" for creatine supplementation and its potential applications for brain health and function. Sports Medicine, Ahead of Print: 1-17, 2023.

Abstract - There is emerging interest regarding the potential beneficial effects of creatine supplementation on indices of brain health and function. Creatine supplementation can increase brain creatine stores, which may help explain some of the positive effects on measures of cognition and memory, especially in aging adults or during times of metabolic stress (i.e., sleep deprivation). Furthermore, creatine has shown promise for improving health outcome measures associated with muscular dystrophy, traumatic brain injury (including concussions in children), depression, and anxiety. However, whether any sex- or age-related differences exist in regard to creatine and indices of brain health and function is relatively unknown. The purpose of this narrative review is to: (1) provide an up-to-date summary and discussion of the current body of research focusing on creatine and indices of brain health and function and (2) discuss possible sex- and age-related differences in response to creatine supplementation on brain bioenergetics, measures of brain health and function, and neurological diseases.

Key Points

- Long-term high-dosage creatine supplementation increases brain creatine stores.
- Creatine supplementation can improve cognition and memory, especially in older adults or during times of metabolic stress (i.e., sleep deprivation).
- Creatine supplementation improves aspects of recovery from traumatic brain injury in children and has the potential to reduce symptoms of depression and anxiety.
- There is some evidence that creatine supplementation improves outcome measures in those with muscular dystrophy but not other neurological diseases or conditions such as Parkinson's disease or amyotrophic lateral sclerosis.

Sports Science Research Update

Myostatin inhibitory peptides in sports drug testing. Drug Testing and Analysis, Ahead of Print: 1-11, 2023.

Abstract - Across species, skeletal muscle mass is negatively regulated by the TGF-β cytokine myostatin (MSTN). Inhibitors of this growth factor and its signaling pathways are therefore not only promising therapeutics for muscular diseases but also potential performance-enhancing agents in sports. Within this study, protein precipitation and liquid chromatography high-resolution tandem mass spectrometry (LC-HRMS/MS) were employed to develop a detection method for six novel MSTN inhibitory pep tides derived from the regulatory MSTN propeptide and the natural MSTN inhibitor follistatin (FST) from doping control serum samples. The approach was comprehensively characterized and found to allow for a specific detection down to concentrations of 3–9 ng/mL. Moreover, several potential metabolites of the drug candidates referred to as DF-3, DF-25, and Peptide 7 were identified as valuable complementary analytical targets for doping control analytical assays. Overall, the acquired data pave the way for an implementation of MSTN inhibitory peptides into routine sports drug testing. Even though no drug candidate has obtained clinical approval yet, a proactive development of detection assays is of utmost importance to deter athletes from misusing such compounds, which are readily available for research purposes and on the black market.

Sports Science Research Update

Neurodiversity in elite sport: A systematic scoping review. BMJ Open Sports & Exercise Medicine, 9(e001575): 1-9, 2023.

Abstract - The objective of this systematic scoping review is to understand the extent and scope of evidence regarding neurodiversity in elite sport. This systematic scoping review considered epidemiological studies, commentary and viewpoints papers, systematic review and meta-analyses, and any intervention or clinical treatment, management and practice studies in relation to neurodiversity in elite sport. Case studies and grey literature were ineligible for review. Neurodivergence included neurodevelopmental disorders such as autism spectrum disorder, attention-deficit hyperactivity disorder (ADHD) and specific learning disorders. Elite sport was defined as Olympic, Paralympic, national, international, professional and semiprofessional sport. The final 23 studies included in this review comprised 10 observational studies, 4 systematic/narrative reviews, 6 commentary/ position statements and 3 qualitative studies. The literature reflected a major focus on ADHD as a risk factor for concussion and prognosis for postconcussion recovery. Further, there was a focus on the medical management of ADHD, regarding adherence to sporting antidoping regulations. One study focused on the experience of autism in athletes in elite sport settings through qualitative interviews. One study focused on anxiety disorders in elite athletes, with ADHD emerging as a major risk factor. There is a strong rationale for future research to build on the evidence for neurodiversity in elite sport to foster supportive and inclusive elite sporting environments.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ There is community and scholarly shift towards understanding neurodiversity from a strengths-based approach, which recognizes differences in cognition, social learning and other behaviors as variations in the context of human neurodevelopment.

⇒ There is yet to be a synthesis of evidence summarizing the state and scope of evidence relating to neurodiversity in elite sport.

WHAT THIS STUDY ADDS

⇒ The findings of this systematic scoping review indicate that while neurodivergence is present in elite sport, there is very limited research in this space to date.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

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⇒ Practice implications of the evidence gap are that neurodiversity inclusive practices are not evidence informed in sport.

⇒ The needs of neurodivergent athletes may be overlooked and the environmental, social, and emotional needs of neurodivergent athletes are yet to be defined and identified in the scientific literature.

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Plant-based diets and sports performance: A clinical review. Sport Science for Health, June 3: 1-23, 2023.

Purpose: Plant-based diets have become increasingly popular in Western culture. Although studies have examined physiologic health improvements of plant-based diets, there is little data on plant-based diets as it relates to sports performance.

Methods: Clinical review of systematic reviews, randomized trials, prospective and retrospective cohort studies available in English on PubMed and Google Scholar databases utilizing combinations of the search terms "athlete, sport, nutrition, diet, vegan, vegetarian, strength, endurance, health, performance, and exercise."

Results: There are no significant differences in athletic performance in participants maintaining plant-based diets compared to those consuming omnivorous diets specifically related to strength, power, and endurance. Plant-based diets are at higher risk of predisposing individuals to certain micronutrient deficiencies, reduced protein intake, and lower serum creatine and sex testosterone levels, though supplementation may effectively replace the lacking components. Non-plant-based diets are associated with a higher risk of obesity, type II diabetes mellitus, certain malignancies, and cardiovascular disease, though several studies demonstrate that a particular meat- and fish-filled diet may reduce body weight, blood pressure, fat composition, and all-cause mortality.

Conclusion: Though plant-based and mixed omnivorous eaters may vary in their macronutrient and micronutrient intake, disease propensities, and oxygen consumption during exertion, a plant-based diet does not translate into a significantly different physical performance compared to a non-plant-based diet across measures of strength, power, and aerobic/anaerobic performance. For athletes, trainers, nutritionists and dieticians, and physicians, the most important recommendation is to maintain a nutritionally sufficient diet that provides the appropriate levels of vital nutrients and vitamins.

Sports Science Research Update

Programming high-speed and sprint running exposure in football: Beliefs and practices of more than 100 practitioners worldwide. International Journal of Sports Physiology and Performance, Ahead of Print: 1-16, 2023.

Purpose: To gain knowledge on the beliefs and practices of football practitioners applying high-speed and sprint running exposure programming strategies.

Methods: One hundred two football practitioners from 22 different countries participated in a study consisting of a survey including 5 domains: demographic and professional characteristics (Who); importance of high-speed and sprint running exposure for physical-capability development, preparation for competition, and injury-prevention strategies (Why); exposure timing (When); methodological procedures for exposure monitoring and training scheduling (What); and effectiveness of common training practices (How). Data were analyzed using a combination of descriptive statistics, generalized mixed effects, and multinomial logistic regression models.

Results: Data revealed 5 main findings: (1) overall agreement on the importance of exposure for physical-capability development, preparation for competition, and injury-prevention strategies; (2) different exposure timing and selective training scheduling for starting and nonstarting players across typical and congested weeks; (3) lack of consensus on the conceptual constructs defining high-speed and sprint running metrics and the methodological procedures used for monitoring; (4) a probable association between match-related outcomes and exposure strategies used in training; and (5) a broad range of training methods considered effective to elicit exposure.

Conclusions: This study provides actionable insights into the planning, implementing, and monitoring strategies for high-speed and sprint running exposure in football. While some conform with the evidence on high-speed and sprint running training in football, further research and professional debate are warranted to develop empirical knowledge and provide pragmatic recommendations to help practitioners in adopting evidence-informed decisions.

Sports Science Research Update

Editorial – Psychological readiness to return to sport: Fear of reinjury is the leading reason for failure to return to competitive sport and is modifiable. Arthroscopy: The Journal of Arthroscopic and Related Surgery, Vol. 39(8): 1775-1778, 2023.

Abstract - Many of our patients are athletes, and an ultimate goal is to help these athletes return to their preinjury level of activity or sport. Generally, we focus on patients' injuries and treatments, but there are modifiable factors that can improve patient outcomes independent of surgical technique. A factor frequently ignored is psychological readiness to return to sport. In patients including athletes, particularly teenagers, chronic, clinical depression is a prevalent and pathologic condition. In addition, in nondepressed patients (or in patients who are situationally depressed because of injury), the ability to deal with stressors may still govern clinical outcomes. Specific psychological traits of significance have been identified and defined, including self-efficacy; locus of control; resilience; catastrophizing; kinesiophobia; and fear of reinjury. (Fear of reinjury is the leading reason for failure to return to competitive sport, reduced activity levels after sports injury, and greater reinjury rates.) The traits may overlap and may be modifiable. Thus, like strength and functional testing, we should evaluate for signs or symptoms of depression, and we should measure psychological readiness to return to sport. With awareness, we can intervene or refer as indicated. Psychological readiness to return to sport represents an underexamined domain in which we can help our patients achieve best outcomes.

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Sleep/wake disorders after sports concussion: Risks, revelations, and interventions. Journal of Clinical Neurophysiology, Vol. 40(5): 417-425, 2023.

Summary - Sleep-wake disturbances (SWDs) are among the most prevalent, persistent, and often disregarded sequelae of traumatic brain injury. Identification and treatment of SWDs in patients with traumatic brain injury is important and can complement other efforts to promote maximum functional recovery. SWDs can accentuate other consequences of traumatic brain injury, negatively affect mood, exacerbate pain, heighten irritability, and diminish cognitive abilities and the potential for recovery. The risk for sports injuries increases when athletes are sleep deprived. Sleep deprivation increases risk-taking behaviors, predisposing to injuries. SWDs are an independent risk factor for prolonged recovery after sports-related concussion. SWDs following sports-related concussion have been shown to impede recovery, rehabilitation, and return to preinjury activities.

In Closing - Sleep-wake disturbances are an independent risk factor for prolonged recovery after SRC. Sleep-wake disturbances after SRC have been shown to impede recovery, rehabilitation, and return to preinjury activities. Identification and treatment of sleep disorders in patients with SRC is important and can complement other efforts to promote maximum functional recovery. SWDs can accentuate other consequences of SRC, negatively affect mood, exacerbate pain, heighten irritability, and diminish cognitive abilities and the potential for recovery. The risk for sports injuries increases when athletes are sleep-deprived. Sleep deprivation increases risk-taking behaviors, predisposing to injuries.